

# Bogyeom Park

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Google Scholar · GitHub

## EDUCATION

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### Seoul National University of Science and Technology

*Ph.D. in Applied Artificial Intelligence*

- Advisor: Kyoungwon Seo

Seoul, South Korea

*Mar. 2023 - Present*

### Seoul National University of Science and Technology

*B.S. in Electrical and Information Engineering*

Seoul, South Korea

*Mar. 2019 - Feb. 2023*

## HONORS & AWARDS

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- AI SeoulTech Graduate Scholarship (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- Best Paper Award (co-author), HCI Korea 2025
- Best Student Paper Bronze Award, IEEE Seoul Section, 2024
- National Science and Engineering Undergraduate Scholarship (full tuition), Korea Student Aid Foundation, 2021-2022
- Best Capstone Design Award, Seoul National University of Science and Technology, 2022

## JOURNAL ARTICLES

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### Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study

**Bogyeom Park**, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

*Journal of Medical Internet Research*, 26, e54538 (2024) - JCR Q1; Top 3%; JIF 8.2 (2025 JCR)

## INTERNATIONAL CONFERENCE PAPERS

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### Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot

**Bogyeom Park** and Kyoungwon Seo

*CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work* (32.7% acceptance; 619/1,888)

### How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment

Minyoung Park, **Bogyeom Park**, and Kyoungwon Seo

*CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work* (32.7% acceptance; 619/1,888)

### A Self-Determination Theory-Based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement

Hyerim Han, **Bogyeom Park**, and Kyoungwon Seo

*CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work* (32.7% acceptance; 619/1,888)

### Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment

**Bogyeom Park**, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

*CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work* (33.9% acceptance; 391/1,154)

### Multimodal Machine Learning Model for MCI Detection Using EEG, MRI and VR Data

Mariem Kallel, **Bogyeom Park**, Kyoungwon Seo, and Seong-Eun Kim

*2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC)*, pp. 1-6

## **Advancing Mild Cognitive Impairment Detection: Integrating VR, MRI, and Neuropsychological Insights for Comprehensive Diagnosis**

**Bogyeon Park**, Jinseok Park, Hojin Choi, Hokyoung Ryu, and Kyoungwon Seo

2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

## **Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning**

Se Young Kim, **Bogyeon Park**, Dohyun Kim, Hojin Choi, Jinseok Park, Hokyoung Ryu, and Kyoungwon Seo

2024 International Conference on Electronics, Information, and Communication (ICEIC), pp. 1-4

## **KOREAN CONFERENCE PAPERS**

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### **From Teacher Needs to Agentic AI: Designing and Validating a Personalized Career Counseling System**

**Bogyeon Park**, Mina Yoo, Dongkuk Lee, Mi-ae Choi, Seona Park, So Young Jo, and Kyoungwon Seo

Proceedings of HCI Korea 2026 - Oral Presentation

### **Counterfactual vs. Prefactual: Two Narrative AIs Improve Causability for Health Data by Different Mechanisms**

**Hyobin Park** (now **Bogyeon Park**) and Kyoungwon Seo

Proceedings of HCI Korea 2023, pp. 828-835

## **RESEARCH EXPERIENCE**

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### **Human-centered Artificial Intelligence (HAI) Lab, SeoulTech**

Seoul, South Korea

#### **Graduate Researcher (Advisor: Kyoungwon Seo)**

Mar. 2023 - Present

#### **Undergraduate Researcher**

Mar. 2021 - Feb. 2023

### **ICAP-Based AI Tutoring System for Probability and Statistics Learning**

Research Assistant | Tutorus Labs

Mar. 2026 - Aug. 2026

- Led the design and evaluation of an ICAP-based AI tutor that uses staged elicitation to promote active, constructive, and interactive engagement rather than answer delivery
- Developed an utterance-level coding and analytics workflow linking dialogue evidence with tutor correctness, usage patterns, and learning outcomes
- Built an interactive research dashboard for reviewing engagement labels and comparing AI-tutored learning processes

### **Agentic AI for Personalized Career, Academic, and Counseling Support**

Research Assistant | Korea Education & Research Information Service

Jun. 2025 - Dec. 2025

- Led a study of how agentic AI could support personalized educational pathways, counseling, and decision-making

### **AI Copilot Technologies for Adaptive, Teacher-Augmented Learning**

Research Assistant | Institute for Information & Communications

Jul. 2023 - Dec. 2025

Technology Planning & Evaluation

- Designed and evaluated a multi-agent LLM debate chatbot that draws out a student's argument and assesses critical thinking from the exchange
- Co-developed a self-determination-theory career counseling chatbot that addresses career decision-making difficulties through motivational dialogue

### **AI-Based Early Screening and Prognosis Prediction for Landscape Tree Disease**

Research Assistant | SUNGHA Co., Ltd.

Jul. 2023 - Dec. 2023

- Built an expert-validated image dataset of *Zelkova serrata*, the species that accounts for more than half of Korea's legally protected trees
- Compared transfer-learning models with and without plant-disease pre-training for early screening of tree disease

### **VR-EP-EEG-MRI Digital Biomarker Basic Research Laboratory**

Research Assistant | National Research Foundation of Korea

Mar. 2022 - Feb. 2024

- Ran the VR data collection on site at Hanyang University Guri Hospital with patients with mild cognitive impairment and healthy controls
- Analyzed the resulting VEEM multimodal dataset - VR interaction, eye tracking, evoked potentials, EEG and MRI - including the VR-MRI-SNSB cohort of 54 participants

## **Multimodal Deep Learning for Early Dementia Diagnosis from VR Daily-Living Data**

Research Assistant | National Research Foundation of Korea

Mar. 2021 - Feb. 2024

- Developed the virtual kiosk task in Unity - the six-step daily-living scenario the study's VR biomarkers are derived from
- Led the development and validation of a multimodal model combining VR behavioral biomarkers with MRI for early detection of mild cognitive impairment
- Derived hand-movement and gaze features from the kiosk task and tested them against MRI-derived measures in a clinical sample

## **TEACHING & MENTORING**

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### **2026 AX Academy Big Data Boot Camp**

May 2026 - Aug 2026

Tutor, Hyundai Motor Group

- Mentored nine participants from across the company, each developing an individual big data project from proposal to final deliverable
- Advised on analysis design and revised project plans with participants through the boot camp's project-based sprints

### **Deep Learning**

Fall 2023

Teaching Assistant, Seoul National University of Science and Technology

- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

## **SKILLS**

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- **Research Methods** - experimental design, usability evaluation, user research, prototyping, multimodal learning analytics, statistical analysis, qualitative coding
- **AI and Data** - machine learning, deep learning, feature engineering, LLM applications, AI agents
- **Programming and Analysis** - Python, C/C#, SPSS
- **VR and Interactive Systems** - Unity, VR task development, interaction logging and eye tracking
- **Design** - HCI theory, UX design, interaction prototyping, usability testing

## **PATENT**

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Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

## **ACADEMIC SERVICE**

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- Student Volunteer, AIED 2026
- Publicity Manager, SeoulTech Human-centered Artificial Intelligence Lab